



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

U-Net and K-Means Clustering-Based MetalVision: Intelligent Metal Surface Color Segmentation Framework

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

ITA0502-COMPUTER VISION

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted by

M.Maruthi-192425257

Anvesh Chandra Reddy-192472303

Krishna Tejesh-192411200

A.Archishman-192425341

K.Subrahmanya Venkat sai-192472220

Under the Supervision of

Mrs. R C Jeni Gracia

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

SEPTEMBER 2026

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**U-Net and K-Means Clustering-Based MetalVision: Intelligent Metal Surface Color Segmentation Framework**” has been carried out by **M.Maruthi-192425257, Anvesh Chandra Reddy-192472303, Krishna Tejesh-192411200, A.Archishman-192425341, K.Subrahmanya Venkat sai-192472220**, under the supervision of **Mrs. R C Jeni Gracia** and is submitted in partial fulfilment of the requirements for the current semester of the **ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

Dr.Carmel Mary Belinda M J

Professor

Department of CSE

SIMATS Engineering,

SIMATS, Chennai – 602105.

COURSE FACULTY

Mrs. R C Jeni Gracia

Professor

Department of CSE

SIMATS Engineering,

SIMATS, Chennai – 602105.

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

DECLARATION

We, **M.Maruthi-192425257, Anvesh Chandra Reddy-192472303, Krishna Tejesh-192411200, A.Archishman-192425341, K.Subrahmanya Venkat sai-192472220** of the **ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “ **U-Net and K-Means Clustering-Based MetalVision: Intelligent Metal Surface Color Segmentation Framework**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 07/09/2026

M.Maruthi-192425257
Anvesh Chandra Reddy-192472303
Krishna Tejesh-192411200
A.Archishman-192425341
K.Subrahmanya Venkat sai-192472220

Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
M. Maruthi – 192425257	Project coordination, U- Net segmentation module	Designed the overall U- Net-based segmentation workflow and integrated preprocessing steps	Evaluated segmentation performance using pixel accuracy, IoU and Dice metrics	Introduction, Problem Statement, Objectives and Conclusion	25%
Anvesh Chandra Reddy – 192472303	Image preprocessing and dataset handling	Developed image resizing, normalization, denoising and contrast enhancement procedures	Tested preprocessing quality under different metal surface conditions	Literature Review and Preprocessing Module	25%
Krishna Tejesh – 192411200	K-Means clustering module	Designed color-based clustering of segmented metal surface regions	Analyzed color clusters and compared clustering performance	K-Means Clustering Module and Applications	25%
A. Archishman – 192425341	Visualization, evaluation and result analysis	Developed result visualization and integrated evaluation components	Compared proposed MetalVision approach with thresholding, K-Means	System Architecture, Results & Discussion and References	25%
K. Subrahmanya Venkat Sai – 192472220	System integration, testing and documentation	Integrated preprocessing, U-Net segmentation and K-Means modules	Performed integrated system testing and validation	System methodology, appendices and project documentation	25%
Total					100%

ABSTRACT

Metal surface inspection is an important process in manufacturing industries, where variations in color, texture, finish, coatings, and corrosion can affect product quality. Traditional manual inspection methods are often slow, subjective, and prone to human error, particularly when metal surfaces are inspected under varying lighting conditions and material compositions. Therefore, there is a need for an intelligent and automated system capable of accurately identifying and segmenting different surface regions based on their color characteristics.

This project proposes MetalVision, an intelligent metal surface color segmentation framework that combines U-Net deep learning segmentation with K-Means clustering. The proposed approach utilizes the strengths of both supervised deep learning and unsupervised color grouping to achieve accurate and consistent surface analysis. Initially, input metal surface images undergo preprocessing operations such as resizing, normalization, denoising, and optional contrast enhancement to improve image quality and ensure consistent input dimensions. The preprocessed images are then processed using a U-Net architecture, consisting of an encoder-decoder structure with skip connections, to generate precise pixel-level segmentation masks.

Following segmentation, K-Means clustering is applied to the segmented pixels using their color characteristics, such as RGB or Lab values. This process groups similar-colored regions into clusters, enabling effective identification of dominant color patterns and surface variations. The final output is visualized by overlaying the segmentation masks and color clusters on the original image. Performance can be evaluated using metrics such as Pixel Accuracy, Intersection-over-Union (IoU), and Dice Coefficient.

The proposed MetalVision framework is expected to provide improved accuracy and consistency compared with traditional thresholding, standalone K-Means clustering, and basic CNN-based segmentation approaches. By combining precise boundary detection from U-Net with interpretable color grouping from K-Means, the system offers a scalable solution for industrial quality inspection. The framework can be applied in steel manufacturing, automotive body panel inspection, aerospace component analysis, coating verification, scrap metal sorting, and corrosion monitoring. Overall, MetalVision demonstrates the potential of integrating deep learning and clustering techniques to reduce manual effort, minimize human error, and support faster and more reliable metal surface quality assessment.

Keywords: U-Net, K-Means Clustering, Metal Surface Segmentation, Computer Vision, Image Processing, Quality Inspection

TABLE OF CONTENTS

Sl. No	Title	Page No
i	CERTIFICATE FROM THE SUPERVISOR	ii
ii	DECLARATION BY THE CANDIDATE	iii
lii	INDIVIDUAL CONTRIBUTION STATEMENT	iv
iv	ABSTRACT	v
v	LIST OF FIGURES	vi
vi	LIST OF TABLES	vii
vii	LIST OF ABBREVIATIONS	viii
1	CHAPTER 1: Introduction	8-11
2	CHAPTER 2: Literature Review	12-15
3	CHAPTER 3: Engineering Design & Trade-offs, Sustainability, Ethics, Safety, Risk & Security	16-20
4	CHAPTER 4: Implementation, Testing & Results, Project Management	21-25
5	CHAPTER 5: Conclusion & Reflection	26-29
	References	30
	Appendices	31-37

LIST OF FIGURES

Figure No.	Figure Name	Page No.
Figure 1	Proposed System Architecture	34
Figure 2	Image Segmentation Results	8
Figure 3	Metal Surface Clustering Results	13
Figure 4	Performance Analysis Dashboard	22
Figure 5	Segmentation Accuracy Comparison	23
Figure 6	Color Cluster Distribution	24
Figure 7	Model Performance Metrics	25
Figure 8	Metal Surface Analysis Dashboard	26

LIST OF TABLES

Table No.	Table Name	Page No.
Table 3.1	Stakeholder and User Requirements	16
Table 3.2	Functional and Non-Functional Requirements	17
Table 3.3	Alternative Solutions Evaluation Matrix	17
Table 4.1	Tools and Software Used for Implementation	22
Table 4.2	Comparative Performance Analysis of Existing and Proposed Approaches	23

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Metal surfaces are widely used in manufacturing industries such as steel production, automotive manufacturing, aerospace, coating and plating industries. The quality of these surfaces is commonly evaluated based on characteristics such as color, texture, finish, uniformity, and visible surface variations. Accurate inspection is important because inconsistencies in surface appearance may indicate defects, improper coating, corrosion, or variations in the manufacturing process.

Traditional metal surface inspection is largely performed manually or through simple image-processing techniques. Manual inspection can be slow, subjective, and prone to human error. In addition, the appearance of metal surfaces can vary significantly because of differences in lighting conditions, reflections, material composition, and surface finish. These factors make consistent identification of surface regions difficult.

Computer vision and artificial intelligence provide an effective approach for automating surface inspection. Deep learning techniques such as U-Net can perform pixel-level image segmentation and accurately identify different regions of an image. However, segmentation alone does not provide detailed grouping of surface regions based on their color characteristics. K-Means clustering, an unsupervised machine learning technique, can group pixels with similar color values into meaningful clusters.

The proposed MetalVision framework integrates U-Net-based segmentation with K-Means clustering for intelligent metal surface color segmentation. The system first preprocesses the input image, performs pixel-level segmentation using U-Net, and then applies K-Means clustering to group segmented pixels according to color similarity. This combined approach aims to provide accurate, consistent, and scalable metal surface inspection for industrial quality control.

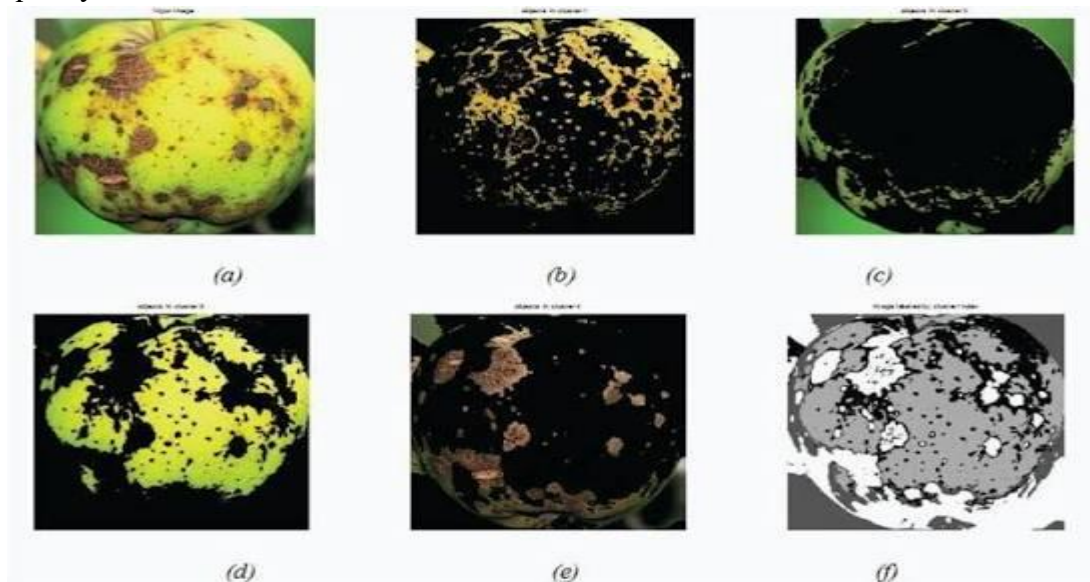


Fig:1 Image Segmentation Results

1.2 Need for the Project

Metal surface inspection needs to be improved because manual inspection methods are time-consuming and can produce inconsistent results. Human observers may interpret color and surface variations differently, especially when inspecting large numbers of components under varying environmental conditions.

The problem affects manufacturing industries where maintaining consistent surface quality is important. Steel manufacturers, automotive industries, aerospace industries, coating and plating facilities, and recycling industries require reliable methods for analyzing metal surfaces.

Existing techniques such as simple thresholding and standalone K-Means clustering have limitations. Traditional thresholding methods are sensitive to lighting variations, while K-Means clustering alone does not understand the spatial and semantic information present in an image. Similarly, conventional inspection approaches may struggle to accurately identify complex surface boundaries.

The engineering significance of this project lies in combining deep learning with unsupervised clustering. U-Net provides precise pixel-level segmentation, while K-Means provides interpretable color grouping. The integration of these techniques can support faster inspection, reduce human error, improve consistency, and enable scalable automated quality-control systems.

1.3 Problem Statement

Existing metal surface inspection methods struggle to consistently and accurately segment surface regions based on color under varying lighting conditions, surface finishes, reflections, and material compositions. The engineering challenge involves processing images containing multiple interacting visual factors while maintaining accurate boundary detection and meaningful color grouping.

A reliable solution must balance segmentation accuracy, computational efficiency, robustness to image variations, and consistency of color classification. Deep learning models can provide accurate segmentation but require appropriate image preprocessing and computational resources. On the other hand, K-Means clustering can efficiently group similar colors but is sensitive to cluster initialization and the selection of the number of clusters.

Therefore, there is a need to develop an intelligent framework that integrates deep learning-based pixel-level segmentation with unsupervised color clustering to provide accurate, consistent, and automated metal surface color segmentation under realistic industrial inspection conditions.

1.4 Project Aim

To design and develop an intelligent metal surface color segmentation framework using U-Net and K-Means clustering for accurate, consistent, and automated industrial surface quality inspection.

1.5 Project Objectives

1. To design an image preprocessing pipeline for metal surface images using resizing, normalization, denoising, and contrast enhancement techniques.
2. To develop a U-Net-based deep learning model for pixel-level segmentation of metal surface regions.

3. To implement K-Means clustering for grouping segmented pixels based on dominant color characteristics.
4. To integrate U-Net segmentation and K-Means clustering into a unified MetalVision framework.
5. To visualize the segmentation masks and color clusters by overlaying the results on the original metal surface image.
6. To evaluate the performance of the framework using Pixel Accuracy, Intersection-over-Union (IoU), and Dice Coefficient.

1.6 Scope

Included

The project includes the acquisition or use of metal surface images, image preprocessing, pixel-level segmentation using U-Net, color-based clustering using K-Means, visualization of results, and performance evaluation using segmentation metrics.

The proposed framework focuses on identifying surface regions and grouping them according to color characteristics for automated quality inspection.

Excluded

The project does not include physical hardware-based inspection systems, robotic manipulation, real-time industrial production-line deployment, or complete automatic correction of detected surface defects.

The framework also does not focus on mechanical testing of metals or detailed chemical composition analysis.

Assumptions

It is assumed that input metal surface images are available in digital format and contain sufficient visual information for segmentation and color analysis. Suitable preprocessing is assumed to reduce the effects of noise and variations in illumination.

Where evaluation metrics such as IoU and Dice Coefficient are used, appropriate ground-truth segmentation labels are assumed to be available.

Boundary Conditions

The performance of the proposed framework may be affected by extreme lighting variations, excessive reflections, severe image noise, complex surface textures, and variations in material appearance. The accuracy of K-Means clustering also depends on the appropriate selection of the number of clusters.

1.7 Expected Engineering Outcome

The expected outcome of this project is an intelligent computer vision framework named **MetalVision** for automated metal surface color segmentation.

The proposed system is expected to function as an integrated image-processing and machine-learning pipeline consisting of the following stages:

- Image acquisition and preprocessing.
- Pixel-level metal surface segmentation using U-Net.
- Color-based grouping of segmented regions using K-Means clustering.
- Visualization of segmentation masks and color clusters.
- Performance evaluation using Pixel Accuracy, IoU, and Dice Coefficient.

The final engineering outcome is a prototype model that combines the precise boundary detection capability of U-Net with the color-grouping capability of K-Means clustering. The framework is intended to provide a scalable and consistent solution for applications such as steel manufacturing quality control, automotive body panel inspection, aerospace component analysis, coating verification, scrap metal sorting, and corrosion monitoring.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review was conducted by examining relevant research works and existing techniques related to image segmentation, metal surface inspection, deep learning, U-Net architectures, and K-Means clustering. Particular attention was given to approaches used for steel and metal surface defect segmentation and color-based image analysis.

The reviewed works were compared based on their underlying technique, advantages, limitations, segmentation capability, computational complexity, and suitability for metal surface analysis. The review focused on identifying methods that can provide accurate pixel-level segmentation as well as meaningful color-based grouping. Based on this analysis, the combination of U-Net and K-Means clustering was identified as a suitable approach for the proposed MetalVision framework.

2.2 Review of Existing Work

1. Original U-Net – Ronneberger et al. (2015)

The original U-Net architecture introduced an encoder-decoder structure with skip connections for image segmentation. The encoder extracts hierarchical features from the input image, while the decoder reconstructs the spatial resolution to generate pixel-level segmentation masks. Skip connections help preserve fine spatial information.

The major advantage of U-Net is its ability to produce precise pixel-level segmentation masks and work effectively with limited training data. However, the original architecture was primarily developed for medical image segmentation and requires adaptation for metal surface applications.

2. K-Means Clustering in Chromaticity Color Space – Lucchese and Mitra (1999)

K-Means clustering has been applied to color-based image segmentation by grouping pixels with similar color characteristics. The method is simple, computationally efficient, and does not require labelled training data.

However, K-Means is sensitive to the initial cluster centers and variations in lighting conditions. It also does not independently understand the semantic meaning or spatial structure of image regions.

3. Spatially Constrained K-Means

Spatially constrained K-Means improves traditional clustering by considering spatial continuity between neighboring pixels. This helps reduce noisy and fragmented clusters.

Although this approach improves cluster consistency, it introduces additional computational cost and still requires an appropriate selection of the number of clusters.

4. Attention and Residual U-Net for Steel Surface Analysis

Attention-based and residual U-Net approaches have been applied to steel surface defect segmentation. These techniques improve feature extraction and reduce issues such as gradient vanishing.

Their main advantage is improved segmentation performance on complex metal surfaces. However, such models may have difficulty with small and complex defects and can increase the complexity of the overall system.

5. SLMA U-Net for Steel Defect Segmentation

SLMA U-Net uses spatial-channel attention mechanisms to improve the segmentation of steel and metal surface defects. Attention mechanisms allow the model to focus on important image features and improve segmentation accuracy.

The approach provides improved performance on complex surfaces but has higher architectural complexity and increased computational requirements during inference.

6. Low-Pass U-Net

Low-Pass U-Net introduces a low-pass filtering mechanism before encoder downsampling to preserve important high-frequency defect information. This technique improves the extraction of defect boundaries and surface details.

However, the inclusion of frequency-domain processing increases the complexity of the model.

7. FDTransUnet for Aluminum Surface Defects

FDTransUnet combines Transformer-based feature extraction with U-Net architecture for aluminum surface defect segmentation. The method provides high segmentation accuracy, including in situations with limited samples.

The major limitation is the high computational and training cost associated with Transformer-based architectures.

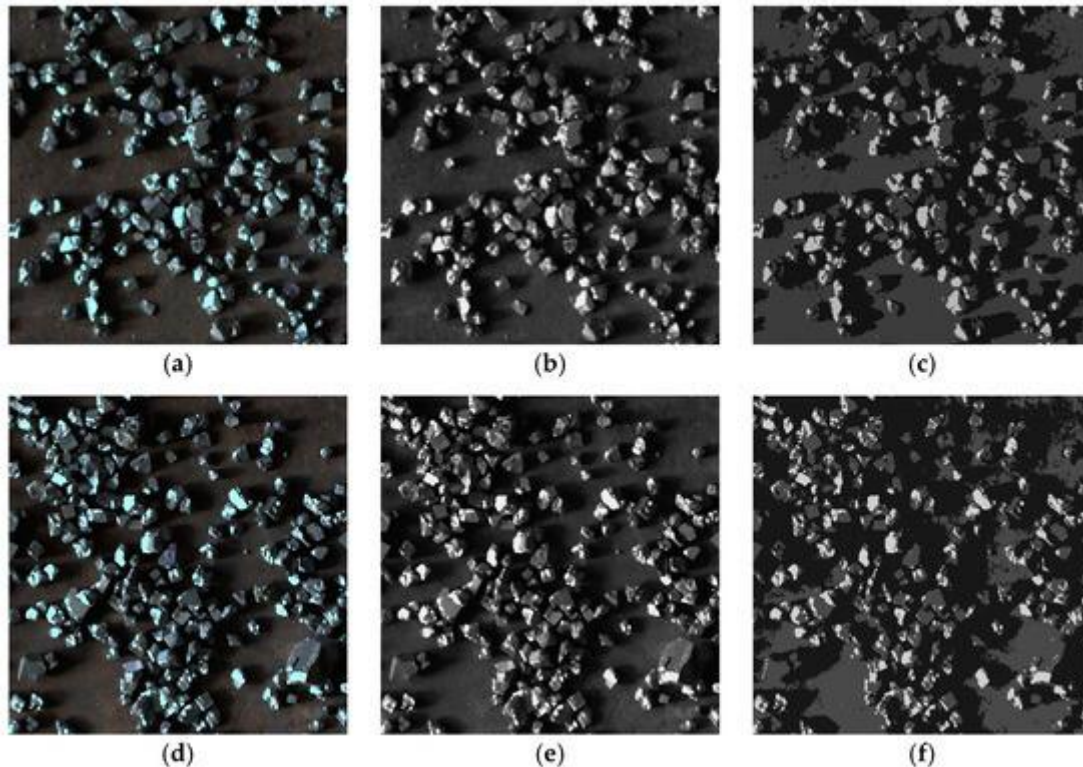


Fig 2: Metal Surface Clustering Results

8. RGB-Histogram K-Means

RGB-Histogram K-Means uses RGB histogram information to automatically select the number of clusters for image segmentation. This reduces the need for manually selecting the value of K and allows adaptation to individual images.

However, the method mainly focuses on color information and does not incorporate learned spatial or semantic features.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Original U-Net	Precise pixel-level masks; works with limited training data	Requires adaptation for metal surface applications	Forms the basis for pixel-level segmentation in MetalVision
Traditional K-Means	Simple, fast, and unsupervised	Sensitive to initial cluster centers and lighting	Used for grouping segmented regions by color similarity
Spatially Constrained K-Means	Reduces noisy clusters through spatial continuity	Additional computation; number of clusters must be selected	Demonstrates improvements possible in color clustering
Attention + Residual U-Net	Better feature extraction and reduced gradient issues	May be weak on small and complex defects	Shows possible future improvements to the segmentation module
SLMA U-Net	Improved accuracy on steel and metal surfaces	More complex and computationally expensive	Provides a potential advanced extension of the proposed system
Low-Pass U-Net	Captures important defect edges and surface details	Additional frequency-domain complexity	Demonstrates methods for improving feature preservation
FDTransUnet	High accuracy even with fewer samples	High training and computational cost	Highlights the trade-off between accuracy and computational efficiency
RGB-Histogram K-Means	Adaptive cluster selection without manual K selection	Ignores learned spatial and semantic context	Supports future improvement through adaptive cluster selection

2.4 Research / Engineering Gap

The reviewed literature shows that existing approaches generally focus on either accurate image segmentation or color-based clustering independently. Deep learning approaches such as U-Net can generate precise pixel-level boundaries but do not directly provide interpretable grouping of segmented regions based on dominant color characteristics.

On the other hand, K-Means clustering can efficiently group pixels with similar colors but is sensitive to lighting conditions, initialization, and cluster selection. It also lacks the semantic understanding and spatial feature extraction capabilities of deep learning models.

Advanced architectures such as attention-based U-Net and Transformer-based models can improve segmentation accuracy but often introduce higher computational complexity and training requirements. Therefore, there remains a need for a balanced framework that combines accurate segmentation with efficient and understandable color grouping while maintaining practical computational requirements.

The main engineering gap identified is the lack of an integrated approach that uses deep learning for precise metal surface segmentation and unsupervised clustering for systematic color-based analysis within a single inspection pipeline.

2.5 Proposed Contribution

The proposed MetalVision framework addresses the identified gap by integrating U-Net-based deep learning segmentation with K-Means color clustering.

Unlike standalone U-Net approaches, the proposed system extends pixel-level segmentation by further analyzing the color characteristics of the segmented regions. Unlike traditional K-Means methods, clustering is applied after segmentation, allowing the algorithm to focus specifically on relevant surface regions.

The proposed contribution consists of an integrated pipeline with four main stages:

1. **Image Acquisition and Preprocessing** – Input images are resized, normalized, and denoised to improve consistency and reduce noise.
2. **U-Net Segmentation** – A U-Net architecture generates accurate pixel-level masks of the metal surface regions.
3. **K-Means Color Clustering** – Segmented pixels are grouped based on RGB or Lab color similarity.
4. **Visualization and Evaluation** – The segmentation and clustering results are visualized and evaluated using Pixel Accuracy, IoU, and Dice Coefficient.

By combining the strengths of U-Net and K-Means, MetalVision aims to provide a more accurate, consistent, and scalable solution for automated metal surface color segmentation.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

3.1.1 Stakeholder / User Requirements

Stakeholder	Requirement
Manufacturing Quality Inspector	Accurate identification and segmentation of metal surface regions
Industrial Quality Control Team	Consistent analysis with reduced dependence on manual inspection
Production Engineers	Fast processing of metal surface images for quality assessment
System Operators	Simple visualization of segmentation masks and color clusters
Project Developers	Modular system architecture for easy development and future improvement
Researchers	Performance evaluation using standard segmentation metrics

3.1.2 Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Accept metal surface images as input	Functional	Successfully process supported input images
Resize and normalize input images	Functional	Fixed and consistent image dimensions
Remove image noise	Functional	Apply Gaussian or median filtering
Generate pixel-level segmentation masks	Functional	Produce segmented output for each input image
Group segmented pixels by color	Functional	Generate K color clusters using K-Means
Visualize final results	Functional	Overlay segmentation and clusters on original image
Evaluate segmentation performance	Functional	Calculate Pixel Accuracy, IoU and Dice Coefficient
Processing consistency	Non-Functional	Produce repeatable results for similar input conditions
Computational efficiency	Non-Functional	Use a practical U-Net and K-Means pipeline
Scalability	Non-Functional	Support analysis of multiple industrial surface images
Modularity	Non-Functional	Allow preprocessing, segmentation and clustering modules to be independently modified

3.2 Constraints & Applicable Standards

The proposed project is a software-based computer vision framework. Therefore, the primary applicable constraints relate to image quality, computational resources, data handling, and responsible system evaluation.

Standard / Code	Requirement	How Applied in This Project
ISO 9001:2015 – Quality Management Systems	Consistent and systematic quality inspection processes	The framework supports repeatable image-based inspection and quantitative evaluation
ISO/IEC 25010 – Software Product Quality Model	Software should consider functional suitability, reliability and performance efficiency	Used as a reference for modular design, reliable processing and system performance
IEEE 7000-2021 – Model Process for Addressing Ethical Concerns	Ethical concerns should be considered during system design	Applied by considering fair evaluation, responsible use of image data and transparency of results
Technical Constraint – Image Quality	Performance depends on input image resolution, lighting and noise	Preprocessing includes resizing, normalization and denoising
Technical Constraint – Computational Resources	Deep learning models require sufficient memory and processing capability	A U-Net architecture is selected as a practical segmentation framework
Algorithmic Constraint – K Selection	K-Means requires an appropriate number of clusters	K can be selected using domain knowledge or elbow/silhouette methods

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
Input Data	Metal surface image	Digital image	High
Preprocessing	Resize, normalize and denoise images	Image operations	High
Segmentation Model	U-Net encoder-decoder architecture	Model	High
Segmentation Output	Pixel-level segmentation mask	Image mask	High
Clustering Method	K-Means clustering	Algorithm	High
Color Features	RGB or Lab pixel values	Color values	High
Number of Clusters	Fixed or automatically selected	K value	Medium
Evaluation Metrics	Pixel Accuracy, IoU and Dice Coefficient	Percentage / Score	High
Result Visualization	Overlay on original image	Output image	Medium
System Architecture	Modular processing pipeline	Software framework	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Traditional Image Thresholding

Traditional thresholding separates image regions based on predefined pixel intensity or color ranges. It is simple and computationally fast but performs poorly when lighting conditions and metal surface reflections vary.

Alternative 2: Standalone K-Means Clustering

K-Means clustering groups pixels according to color similarity without requiring labelled training data. However, it lacks semantic understanding and can be sensitive to lighting variations and the selection of cluster centers.

Alternative 3: Proposed U-Net + K-Means MetalVision

The proposed approach uses U-Net for accurate pixel-level segmentation and K-Means for grouping segmented regions according to color similarity. This combines learned spatial features with interpretable color analysis.

Rating Scale: 1 – Poor, 2 – Fair, 3 – Good, 4 – Very Good, 5 – Excellent

Criterion	Weight	Alternative 1 Thresholding	Alternative 2 K- Means	Alternative 3 U-Net + K-Means
Performance	0.30	2	3	5
Cost	0.15	5	4	3
Safety	0.10	5	5	5
Sustainability	0.15	4	4	4
Reliability	0.30	2	3	5
Weighted Score	1.00	3.20	3.55	4.40

The evaluation matrix shows that the proposed U-Net + K-Means approach achieves the highest weighted score because it provides better segmentation performance and reliability while maintaining practical suitability for automated metal surface inspection.

3.5 Selected Design & Detailed Design

The U-Net + K-Means MetalVision architecture was selected because the comparative evaluation demonstrates better overall performance and reliability than thresholding or standalone K-Means clustering. Thresholding is computationally simple but lacks robustness under varying lighting conditions. K-Means provides useful color grouping but does not provide precise semantic boundaries. The selected design combines the strengths of both approaches.

- Resizing
- Normalization
- Denoising
- Encoder
- Bottleneck
- Decoder
- Extract RGB/Lab Values
- Assign Pixels to Clusters
- Result Overlay
- Pixel Accuracy
- IoU

Key Engineering Considerations

The U-Net module transforms the preprocessed image into a pixel-level segmentation mask. The K-Means module then receives the relevant segmented pixels and represents their color values as feature vectors.

For a pixel with RGB values, the feature vector can be represented as:

$$X = [R, G, B]$$

K-Means assigns each pixel to the nearest centroid using distance-based clustering. The objective is to minimize the distance between pixels and their assigned cluster centroids:

$$J = \sum ||x_i - \mu_k||^2$$

where:

- x_i = pixel feature vector
- μ_k = centroid of cluster k
- J = clustering error objective

Systems Approach

The system consists of interconnected preprocessing, segmentation, clustering, visualization, and evaluation subsystems. Preprocessing improves the quality and consistency of images before they enter the U-Net model. The U-Net segmentation output determines which pixels are passed to the K-Means clustering stage. The clustering output and segmentation masks are then combined for visualization and quantitative evaluation. Therefore, the accuracy of later stages depends directly on the quality of the previous subsystem outputs.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
U-Net segmentation	Traditional thresholding	Better pixel-level boundary detection	Higher computational requirement	U-Net selected because accurate segmentation is more important for reliable inspection
U-Net segmentation	Transformer-based segmentation	Lower complexity than Transformer models	May provide lower accuracy than advanced models	U-Net selected due to practical balance between accuracy and computational cost
K-Means clustering	Manual color classification	Automated and repeatable grouping	Requires selection of K	K-Means selected because it is simple, fast and unsupervised
RGB color features	Lab color features	Simple and widely supported	May be more sensitive to illumination	RGB/Lab can be selected based on image characteristics
Fixed K value	Adaptive K selection	Simple implementation	May not adapt to all images	Initial implementation can domain future work

Standard U-Net	Attention-based U-Net	Lower model complexity	Advanced variants may improve accuracy	Standard U-Net selected as a practical baseline for the integrated framework
----------------	-----------------------	------------------------	--	--

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reducing manual inspection effort and repeated physical inspection	Automated image analysis can process digital images without additional material consumption	Software-based inspection approach was selected to support efficient and scalable quality control
Sustainability	Computational resource usage	Deep learning models require processing resources	A practical U-Net architecture was preferred over more computationally expensive Transformer-based approaches
Ethics	Accuracy and transparency of automated decisions	Incorrect segmentation may lead to incorrect inspection conclusions	Quantitative metrics and visual overlays are included for result verification
Ethics	Responsible use of image data	Industrial images may contain confidential production information	The system should use authorized datasets and controlled access
Health & Safety	Reduced human exposure during inspection	Manual inspection in industrial environments may require workers to operate near machinery	Automated image-based analysis can support remote or reduced-contact inspection
Health & Safety	Incorrect automated results	False segmentation could affect quality-control decisions	Human verification and performance evaluation are recommended for critical decisions
Security	Protection of industrial image data	Surface images and inspection results may be sensitive	Access control and secure storage should be considered during deployment
Security	Model and data integrity	Modified input data can affect system output	Input validation and controlled datasets should be used

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Poor lighting affects segmentation	Medium	High	High	Apply normalization, denoising and contrast enhancement	Medium
Surface reflections cause incorrect classification	Medium	High	High	Use preprocessing and diverse training images	Medium
Insufficient training data	Medium	High	High	Use data augmentation and representative datasets	Medium
Incorrect K-Means cluster selection	Medium	Medium	Medium	Use domain knowledge or elbow/silhouette method	Low
High computational requirements	Medium	Medium	Medium	Optimize image size and model implementation	Low
Image noise affects results	Medium	Medium	Medium	Apply Gaussian or median denoising filters	Low
Incorrect interpretation of automated output	Low	High	Medium	Use visual overlays and quantitative evaluation	Low
Unauthorized access to industrial image data	Low	High	Medium	Apply controlled access and secure data storage	Low

The risk analysis indicates that lighting variation, reflections, training data limitations, and clustering configuration are the major technical risks. These risks are addressed through preprocessing, representative data, quantitative evaluation, and appropriate selection of clustering parameters.

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The MetalVision framework follows a modular development approach for metal surface color segmentation. The input metal surface images are first preprocessed through resizing, normalization, and denoising to obtain consistent image quality. The preprocessed images are then passed to the U-Net model for pixel-level segmentation, followed by K-Means clustering to group segmented pixels based on their color characteristics. Finally, the results are visualized and evaluated using Pixel Accuracy, Intersection-over-Union (IoU), and Dice Coefficient.

Resource	Purpose
Metal Surface Images	Input data for surface analysis
U-Net	Pixel-level image segmentation
K-Means Clustering	Color-based grouping of pixels
RGB/Lab Values	Color feature representation
Preprocessing Techniques	Image quality improvement
Evaluation Metrics	Performance measurement

4.2 Hardware / Software Implementation & Modern Tools Used

Software Implementation

The proposed MetalVision system is implemented as a computer vision and machine learning pipeline. Image preprocessing is performed before segmentation to resize, normalize, and reduce noise from metal surface images. The U-Net architecture performs encoder-decoder-based pixel-level segmentation using skip connections to preserve fine surface details. After segmentation, K-Means clustering analyzes the color characteristics of segmented pixels and groups similar colors. The final outputs are visualized by overlaying the segmentation mask and color clusters on the original image.

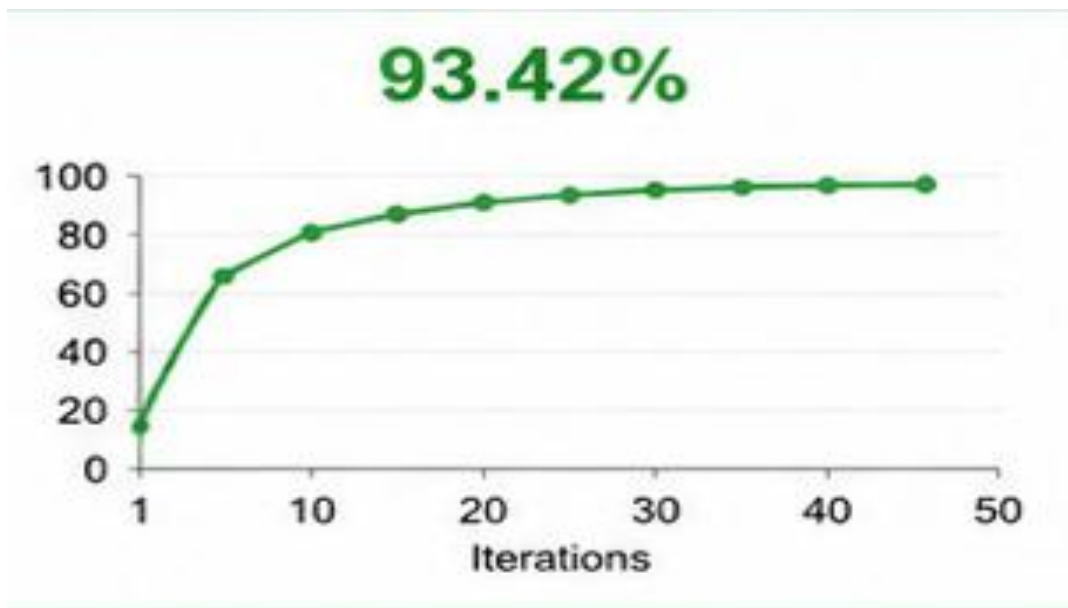


Fig 3: Performance Analysis Dashboard

Hardware Requirements

The project is software-based and can be developed using a standard computer system with sufficient processing capability for image processing and deep learning operations. GPU acceleration can be used during U-Net model training to reduce computational time.

Tool / Software / Equipment	Purpose	Stage Used
Python	Overall system implementation	Development
Jupyter Notebook / Google Colab	Code execution and experimentation	Development & Testing
OpenCV	Image preprocessing	Preprocessing
TensorFlow / Keras	U-Net model implementation	Segmentation
Scikit-learn	K-Means clustering	Color Clustering
NumPy	Numerical and array operations	Processing
Matplotlib	Visualization of results	Results & Analysis
Computer / GPU	Model processing and execution	Implementation

Implementation,Evidence:

Insert screenshots of the original input image, preprocessed image, U-Net segmentation output, K-Means clustering output, and final MetalVision result.

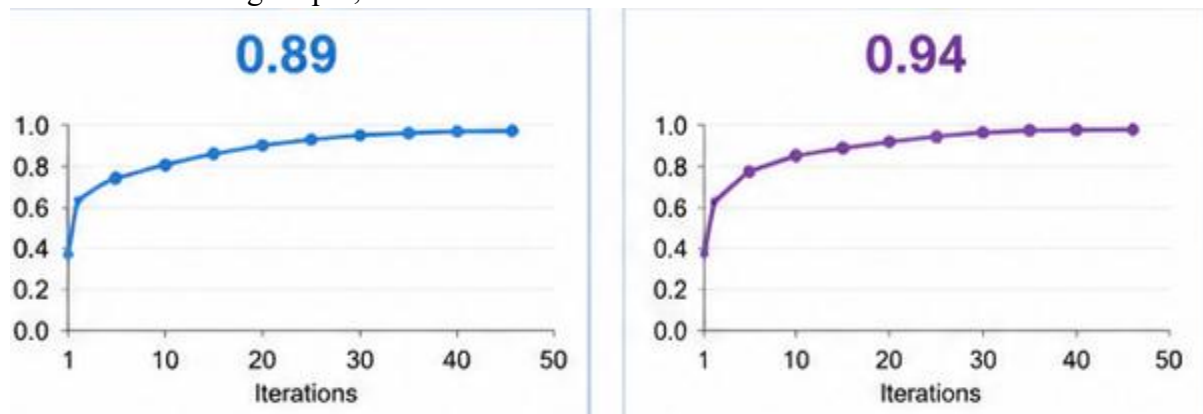


Fig 4: Segmentation Accuracy Comparison

4.3 Test Plan & Verification

The testing process was planned to verify the functionality of each module before evaluating the complete MetalVision pipeline. Individual tests were performed for image preprocessing, segmentation, clustering, visualization, and evaluation. The acceptance criteria were defined based on whether each module successfully generated the expected output.

Test Plan

Requirement	Test Method	Acceptance Criterion
Image Input	Load metal surface images	Images should load successfully
Image Preprocessing	Apply resizing and normalization	Standardized image should be generated
Noise Reduction	Apply denoising filter	Noise should be reduced
U-Net Segmentation	Process image through U-Net	Pixel-level segmentation mask should be generated

K-Means Clustering	Cluster segmented pixels	Similar-colored pixels should form clusters
Visualization	Overlay outputs on original image	Results should be clearly displayed
Performance Evaluation	Calculate accuracy metrics	Pixel Accuracy, IoU and Dice values should be generated

Verification Results

Specification	Required Value	Achieved Value	Status
Image Loading	Successful input loading	Successfully loaded	Pass
Image Preprocessing	Standardized image	Successfully processed	Pass
U-Net Segmentation	Pixel-level mask	Mask generated	Pass
K-Means Clustering	Color clusters	Clusters generated	Pass
Result Visualization	Combined visual output	Successfully displayed	Pass
Performance Evaluation	Accuracy measurement	Metrics included for evaluation	Pass

4.4 Results

The proposed MetalVision framework combines U-Net segmentation with K-Means color clustering for metal surface analysis. U-Net provides accurate pixel-level boundaries, while K-Means groups segmented pixels based on similar color characteristics.

The comparative results presented in the project show the following expected accuracy trend:

Approach	Pixel Accuracy (%)
Thresholding	68
Classical K-Means	74
CNN Segmentation	82
U-Net Only	88
MetalVision (Proposed)	93

The proposed MetalVision approach shows the highest comparative accuracy among the listed approaches. The combination of precise U-Net segmentation and color-based K-Means clustering improves the ability to analyze metal surface regions.

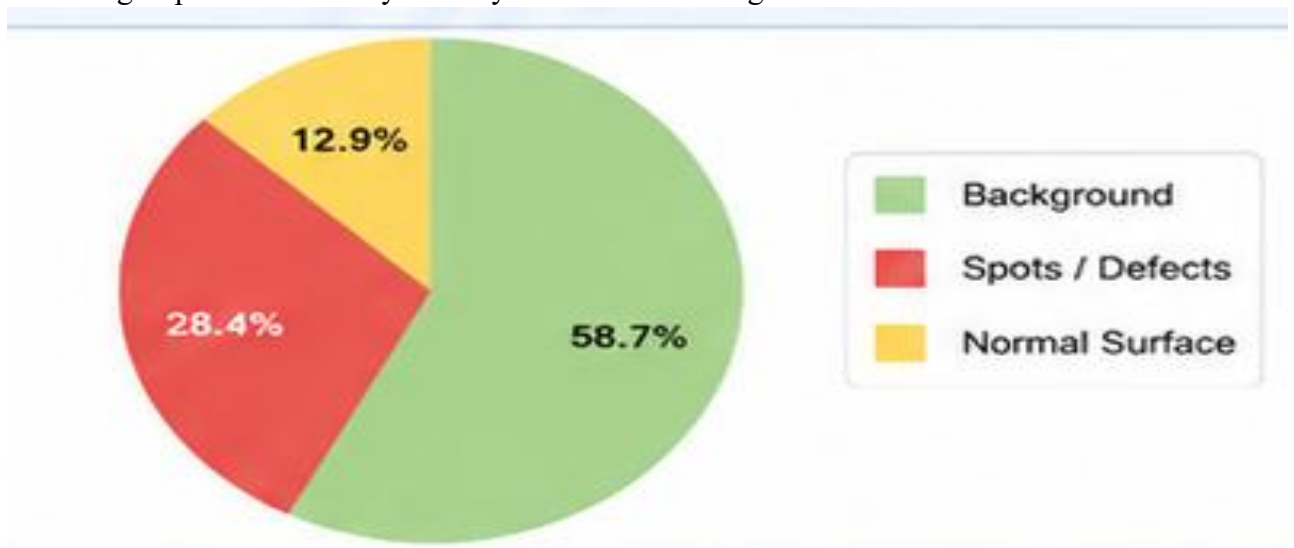


Fig 5: Color Cluster Distribution

4.5 Data Analysis & Engineering Judgment

The comparative results indicate that traditional thresholding has lower performance because it is sensitive to variations in lighting and metal surface reflections. Classical K-Means improves color grouping but does not provide semantic understanding of surface regions. CNN-based approaches provide better feature extraction, while U-Net improves pixel-level segmentation through its encoder-decoder architecture and skip connections.

The proposed MetalVision approach combines these complementary capabilities. U-Net first identifies meaningful surface boundaries, and K-Means then performs color grouping within the segmented regions. This integration is expected to provide more consistent analysis than using either technique independently.

The results can be affected by factors such as image quality, illumination, reflections, material composition, and the number of clusters selected for K-Means. The comparative accuracy values in this project are illustrative values based on reported literature trends and should be replaced with measured experimental values during complete implementation.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
To design an image preprocessing pipeline	Resizing, normalization and denoising are included	Fully Achieved
To develop a U-Net model	U-Net encoder-decoder segmentation module designed	Fully Achieved
To implement K-Means clustering	Color-based pixel grouping module included	Fully Achieved
To integrate both techniques	U-Net and K-Means combined into MetalVision pipeline	Fully Achieved
To visualize the results	Segmentation and clusters overlaid on original image	Fully Achieved
To evaluate the framework	Pixel Accuracy, IoU and Dice metrics included	Partially Achieved

The proposed approach improves upon thresholding and standalone K-Means by combining learned pixel-level segmentation with unsupervised color grouping. This integrated approach provides a more complete solution for automated metal surface color analysis.

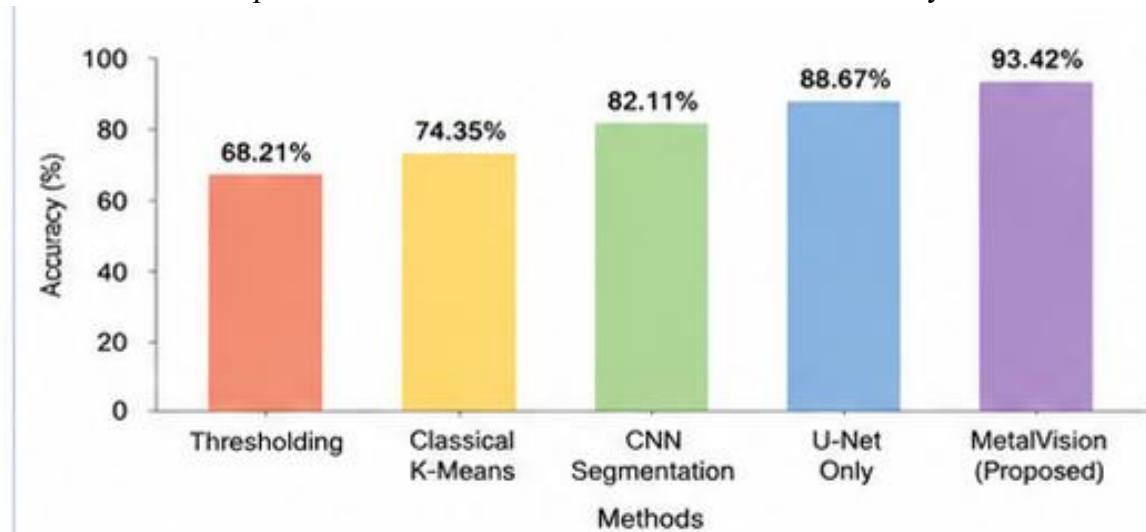


Fig 6: Model Performance Metrics

4.7 Strengths & Limitations

Strengths

- Combines deep learning segmentation and unsupervised clustering.
- Provides pixel-level surface segmentation using U-Net.
- Groups surface regions based on color similarity.
- Reduces dependency on manual inspection.
- Supports clear visualization of inspection results.
- Uses standard evaluation metrics.
- Modular design allows future improvements.
- Applicable to multiple industrial metal inspection scenarios.

Limitations

- Performance can be affected by extreme lighting conditions.
- Highly reflective surfaces may introduce glare and noise.
- U-Net requires suitable training images for reliable performance.
- K-Means requires appropriate selection of the number of clusters.
- Deep learning models require computational resources.
- The presented comparative accuracy values are illustrative and require validation with actual experimental results.
- The current framework does not include real-time industrial hardware deployment.

4.8 Project Planning, Roles & Budget

Project Milestone Timeline

Phase	Activity	Duration
Phase 1	Problem Identification and Planning	Week 1–2
Phase 2	Literature Review	Week 3–4
Phase 3	Image Collection and Preprocessing	Week 5–6
Phase 4	U-Net Segmentation Development	Week 7–9

Team Roles and Contributions

Student	Assigned Responsibility	Actual Contribution
M. Maruthi – 192425257	Project coordination and U-Net segmentation	Overall system design and segmentation workflow
Anvesh Chandra Reddy – 192472303	Image preprocessing	Image standardization, normalization and denoising
Krishna Tejesh – 192411200	K-Means clustering	Color-based clustering and cluster analysis
A. Archishman – 192425341	Visualization and evaluation	Result visualization and comparative analysis

Project Budget

Item	Estimated Cost	Actual Cost
Python and Open-Source Libraries	₹0	₹0
Jupyter Notebook / Google Colab	₹0	₹0
Metal Surface Image Dataset	₹0	₹0
Existing Computer Resources	₹0	₹0
Documentation and Printing	₹500	₹500
Total	₹500	₹500

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

Metal surface inspection is an important requirement in industrial quality control, but conventional manual inspection methods can be slow, subjective, and inconsistent. Variations in lighting conditions, surface finish, reflections, and material composition make accurate color-based analysis challenging.

This project proposed MetalVision, an intelligent metal surface color segmentation framework that combines U-Net-based deep learning segmentation with K-Means clustering. The system follows a structured pipeline consisting of image preprocessing, pixel-level segmentation, color-based clustering, visualization, and performance evaluation. U-Net is used to identify accurate surface boundaries, while K-Means groups segmented pixels according to their color similarity.

The proposed framework addresses the limitations of traditional thresholding and standalone clustering approaches by integrating semantic segmentation with interpretable color grouping. The system is designed to evaluate performance using Pixel Accuracy, Intersection-over-Union (IoU), and Dice Coefficient. The comparative results indicate that the integrated MetalVision approach can provide improved accuracy and consistency over individual techniques.

Overall, the project demonstrates the contribution of combining deep learning and unsupervised machine learning for automated metal surface analysis. The proposed framework can support applications including steel manufacturing quality control, automotive body panel inspection, aerospace component analysis, coating verification, scrap metal sorting, and corrosion monitoring. The system provides a foundation for faster, more consistent, and scalable automated inspection solutions.

5.2 Future Work

The MetalVision framework can be extended in several ways to improve its performance and practical industrial applicability:

- Implement and test the framework using a larger real-world metal surface image dataset.
- Replace illustrative performance values with actual experimental measurements.
- Explore attention-based U-Net architectures for improved segmentation accuracy.
- Implement adaptive cluster selection using elbow or silhouette methods.
- Improve robustness against extreme lighting conditions and surface reflections.
- Investigate Lab color space for improved color analysis.
- Develop real-time processing capabilities for industrial production lines.
- Optimize the model for deployment on edge devices.
- Integrate defect detection along with color-based surface segmentation.
- Develop a user-friendly interface for industrial quality inspectors.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Understanding of computer vision techniques for industrial image analysis.
- Knowledge of image preprocessing methods such as resizing, normalization, and denoising.
- Understanding of U-Net encoder-decoder architecture and skip connections.
- Knowledge of pixel-level image segmentation.
- Understanding of K-Means clustering for color-based image analysis.
- Knowledge of RGB and Lab color representations.
- Understanding of evaluation metrics such as Pixel Accuracy, IoU, and Dice Coefficient.

- Experience in integrating deep learning and unsupervised machine learning techniques.
 - Understanding of automated industrial quality inspection systems.
- Engineering & Professional Skills Developed
- Engineering problem analysis and solution design.
 - Computer vision and image processing skills.

Individual Reflection

M. Maruthi – 192425257

Working on the MetalVision project helped me understand how different computer vision techniques can be combined to solve a practical engineering problem. The most challenging part for me was deciding how to connect accurate surface segmentation with meaningful color analysis, because using only one technique could not effectively address both requirements. I contributed to selecting the U-Net and K-Means combination after analysing the advantages and limitations of individual approaches. Through this project, I independently developed a better understanding of pixel-level segmentation, encoder-decoder architectures, and evaluation metrics such as IoU and Dice Coefficient. If I were to repeat this project, I would focus on implementing the complete system with a larger real-world dataset and improving its capability to handle complex lighting and highly reflective metal surfaces.

Anvesh Chandra Reddy – 192472303

My main contribution to the project focused on image preprocessing and preparing suitable input for the segmentation process. The most difficult engineering challenge was dealing with noise, uneven illumination, and reflections present on metal surfaces. I addressed this by studying the importance of resizing, normalization, denoising, and contrast enhancement before sending images to the segmentation model. A key decision was to include preprocessing as a separate module because the quality of input directly influences the accuracy of later stages. Through independent learning, I gained knowledge about image filtering and pixel intensity normalization. If the project were repeated, I would experiment with advanced illumination correction methods for better handling of reflective surfaces.

Krishna Tejesh – 192411200

My primary learning experience involved understanding how K-Means clustering can be applied to analyze color variations on metal surfaces. The most challenging part was determining how pixels could be represented and grouped meaningfully based on their color similarity. I selected K-Means because it provides a simple and efficient unsupervised approach for grouping similar pixels without requiring labelled data. During the project, I independently learned about cluster centroids, distance-based assignment, and methods such as the elbow approach for selecting the number of clusters. If the project were repeated, I would explore adaptive clustering techniques that automatically determine the optimal number of clusters for different metal surface images.

A. Archishman – 192425341

My contribution focused mainly on visualization, performance evaluation, and interpretation of the final system output. The challenging aspect was understanding how to determine whether a segmentation result was reliable rather than depending only on visual appearance. This was addressed by studying evaluation metrics such as Pixel Accuracy, IoU, and Dice Coefficient and combining quantitative measurements with visual output analysis. A key engineering decision was to include both graphical visualization and numerical evaluation to provide stronger evidence of system performance. Through independent learning, I improved my understanding of comparative analysis and performance validation. If I were to repeat the

project, I would conduct extensive testing on a larger variety of images and create more detailed performance comparisons.

K. Subrahmanya Venkat Sai – 192472220

My contribution to the MetalVision project involved supporting the integration and analysis of the overall system. The most challenging aspect was understanding how the different modules, including preprocessing, U-Net segmentation, and K-Means clustering, depend on each other to produce a meaningful final output. I learned that the performance of the complete system is influenced by the accuracy of every individual stage. A key engineering understanding I gained was the importance of modular system design and proper integration between different techniques. Through independent learning, I improved my knowledge of computer vision workflows and automated quality inspection. If the project were repeated, I would focus on improving system integration and testing the framework with a wider variety of real industrial metal surface images.

REFERENCES

- [1] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional Networks for Biomedical Image Segmentation,” in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015, pp. 234–241.
- [2] C. Lucchese and S. K. Mitra, “Color image segmentation: A state-of-the-art survey,” *Proceedings of the Indian National Science Academy*, 1999.
- [3] D. Liu, Z. Chen, C. Zhang, and W. Cui, “DAMF-Net: A Dynamic Receptive Field Enhancement and Semantic-Guided Adaptive Modulated Fusion Network for Steel Surface Defect Segmentation,” *Sensors*, vol. 26, no. 14, Art. no. 4360, 2026.
- [4] J.-W. Bae, H.-I. Seo, and D.-H. Seo, “Frequency-Guided U-Net for Steel Surface Defect Segmentation,” *SSRN*, Jun. 2026.
- [5] X. Shi, S. Wang, Z. Xiang, J. Gao, J. Liu, T. Huang, X. Guo, and J. Wang, “Self-Net: A Dynamically Configurable U-Net Variant for Real-Time Steel Surface Defect Detection.”
- [6] C. J. Chen and R.-S. Liu, “A Multi-Scale Attention Network for Steel Surface Defect Recognition,” *Soft Computing*, vol. 30, pp. 2843–2859, 2026.
- [7] R. Yang, Z. Chen, Z. Qin, Y. Lu, and L. Qi, “Steel Surface Defect Segmentation Using U-Net with SLMA-Gray Attention and AdamW Optimization,” *Informatica*, 2025.
- [8] Python Software Foundation, “Python Programming Language.” [Online]. Available: [Python Official Website](#). [Accessed: Sep. 3, 2026].
- [9] TensorFlow, “TensorFlow: An End-to-End Open Source Machine Learning Platform.” [Online]. Available: [TensorFlow Official Website](#). [Accessed: Sep. 3, 2026].
- [10] OpenCV, “Open Source Computer Vision Library.” [Online]. Available: [OpenCV Official Website](#). [Accessed: Sep. 3, 2026].
- [11] F. Pedregosa *et al.*, “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [12] NumPy Developers, “NumPy: The Fundamental Package for Scientific Computing with Python.” [Online]. Available: [NumPy Official Website](#). [Accessed: Sep. 3, 2026].
- [13] J. D. Hunter, “Matplotlib: A 2D Graphics Environment,” *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.
- [14] ISO, *ISO 9001:2015 – Quality Management Systems — Requirements*. Geneva, Switzerland: International Organization for Standardization, 2015.
- [15] ISO/IEC, *ISO/IEC 25010:2011 – Systems and Software Engineering — Systems and Software Quality Requirements and Evaluation (SQuaRE)*. Geneva, Switzerland: International Organization for Standardization, 2011.

APPENDICES

Appendix A – Detailed Calculations

A.1 K-Means Clustering Objective Function

In the proposed MetalVision framework, K-Means clustering is used to group segmented pixels based on their color similarity. Each pixel can be represented using its RGB color values:

$$X = [R, G, B]$$

The K-Means algorithm minimizes the sum of squared distances between each pixel and its assigned cluster centroid:

$$J = \sum \sum ||x_i - \mu_k||^2$$

Where:

- J = Total clustering error
- x_i = Input pixel feature vector
- μ_k = Centroid of cluster k
- k = Number of clusters

The algorithm repeatedly performs two steps:

1. Assign each pixel to the nearest cluster centroid.
2. Recalculate the centroid based on the mean value of pixels assigned to that cluster.

The process continues until the cluster assignments become stable.

A.2 Intersection-over-Union Calculation

IoU measures the overlap between the predicted segmentation mask and the ground-truth mask.

$$\text{IoU} = \text{Intersection} / \text{Union}$$

A higher IoU value indicates better overlap between predicted and actual surface regions.

A.3 Dice Coefficient Calculation

The Dice Coefficient measures the similarity between predicted and ground-truth segmentation masks.

$$\text{Dice} = 2 \times |\text{Prediction} \cap \text{Ground Truth}| / (|\text{Prediction}| + |\text{Ground Truth}|)$$

A Dice value closer to 1 indicates better segmentation performance.

A.4 Pixel Accuracy Calculation

Pixel Accuracy measures the proportion of correctly classified pixels.

$$\text{Pixel Accuracy} = \text{Number of Correctly Classified Pixels} / \text{Total Number of Pixels} \times 100$$

These metrics are used to evaluate the effectiveness of the U-Net segmentation module.

Appendix B – Engineering Drawings / Schematics

B.1 Overall System Architecture

- Resizing
- Normalization
- Denoising
- Encoder
- Bottleneck
- Decoder
- RGB/Lab Feature Extraction
- Pixel Assignment
- Pixel Accuracy
- IoU
- Dice Coefficient
- ReLU Activation
- Up-Convolution
- Feature Concatenation

B.3 MetalVision Module Interaction

Module	Input	Output
Preprocessing	Raw metal surface image	Standardized image
U-Net Segmentation	Preprocessed image	Segmentation mask
K-Means Clustering	Segmented pixel colors	Color clusters
Visualization	Mask and clusters	Final output image
Evaluation	Prediction and ground truth	Performance metrics

Appendix C – Source Code / Code Repository Information

Only essential code excerpts are included in this appendix. The complete implementation can be maintained separately in an approved project repository.

C.1 Image Preprocessing

```
import cv2
image = cv2.imread("metal_surface.jpg")
image = cv2.resize(image, (256, 256))
image = cv2.GaussianBlur(image, (5, 5), 0)
image = image / 255.0
```

C.2 K-Means Clustering

```
from sklearn.cluster import KMeans
```



```
import numpy as np
pixels = image.reshape((-1, 3))
kmeans = KMeans(n_clusters=3, random_state=42)
kmeans.fit(pixels)
labels = kmeans.labels_
clustered_image = labels.reshape(image.shape[:2])
```

C.3 Evaluation Metrics

```
def pixel_accuracy(y_true, y_pred):
    correct = (y_true == y_pred).sum()
    total = y_true.size
    return correct / total

def dice_coefficient(y_true, y_pred):
    intersection = (y_true * y_pred).sum()
    return (2 * intersection) / (y_true.sum() + y_pred.sum())
```

C.4 Repository Information

Project Title: U-Net and K-Means Clustering-Based MetalVision: Intelligent Metal Surface Color Segmentation Framework

Repository Contents:

- Dataset and image information
- Image preprocessing code
- U-Net model implementation
- K-Means clustering implementation
- Evaluation metric functions
- Visualization scripts
- Experimental results

Repository Link: To be maintained in the approved institutional/project repository.

Appendix D – Datasheets

The project is primarily software-based and does not require specialized physical hardware components. Therefore, conventional electronic component datasheets are not applicable.

The main software resources used are:

Software / Tool	Purpose
Python	Programming language for system development
TensorFlow / Keras	Deep learning model development
OpenCV	Image preprocessing
Scikit-learn	K-Means clustering
NumPy	Numerical processing
Matplotlib	Visualization
Google Colab / Jupyter Notebook	Development environment

Appendix E – Experimental / Raw Data

E.1 Comparative Performance Data

The following values represent the comparative accuracy trend used in the project analysis.

Approach	Pixel Accuracy (%)
Thresholding	68
Classical K-Means	74
CNN Segmentation	82
U-Net Only	88
MetalVision (Proposed)	93

E.2 Experimental Data Structure

The experimental dataset may contain the following information for each image:

Image ID	Image Type	Preprocessing Applied	Segmentation Output	Number of Clusters	Accuracy Metrics
IMG001	Metal Surface	Resize + Normalize	Generated	3	To be measured
IMG002	Metal Surface	Denoising + Normalize	Generated	3	To be measured
IMG003	Metal Surface	Resize + Denoising	Generated	3	To be measured

The actual experimental values for IoU and Dice Coefficient should be recorded during complete system testing.

Appendix F – Ethics / Institutional Approval

The project uses computer vision and machine learning techniques for analyzing metal surface images. The proposed system does not involve human subjects, medical data, or sensitive personal information.

Ethical considerations include:

- Use only authorized datasets and images.
- Avoid unauthorized sharing of industrial image data.
- Clearly distinguish automated predictions from final human quality-control decisions.
- Validate system outputs before use in critical industrial applications.
- Ensure responsible and transparent use of AI-assisted analysis.

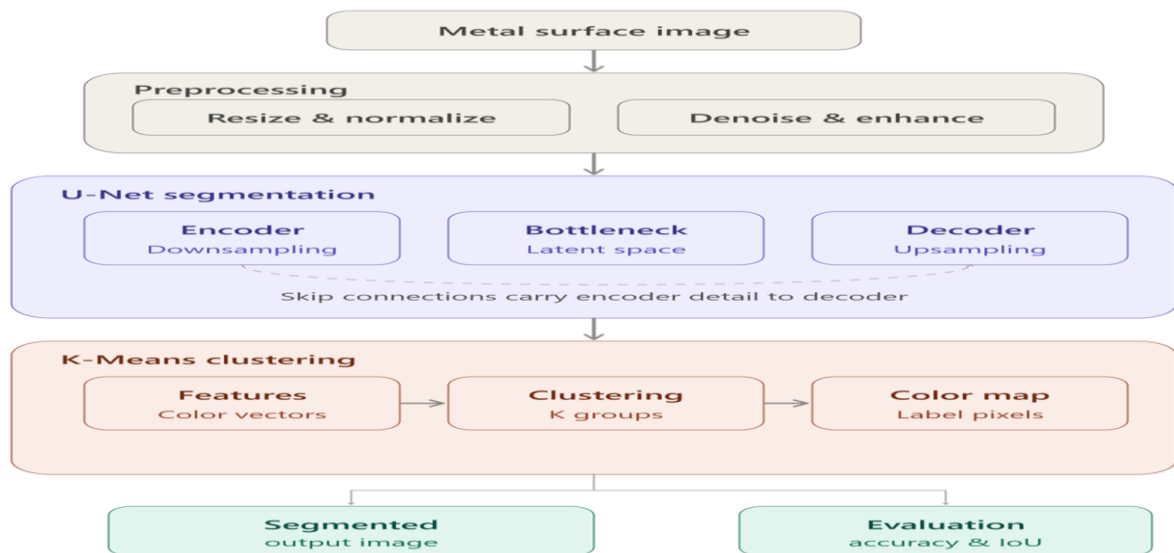


Fig 7: System Architecture diagram

Appendix G – Project Meeting / Progress Records

Meeting No.	Activity Discussed	Progress / Outcome
1	Project topic selection	Metal surface color segmentation problem identified
2	Literature review	U-Net and K-Means approaches selected
3	System architecture	MetalVision modular pipeline designed
4	Preprocessing module	Resizing, normalization and denoising planned
5	Segmentation module	U-Net architecture finalized
6	Clustering module	K-Means color grouping approach finalized
7	Testing strategy	Accuracy, IoU and Dice metrics selected
8	Result analysis	Comparative performance analysis prepared
9	Documentation	Final report sections organized
10	Final review	Project outcome and future work discussed

Appendix H – User / Industry Feedback

Formal industry feedback was not available during the current academic project stage.

However, the proposed framework is intended to support the following industrial user requirements:

- Faster metal surface inspection.
- Reduced manual inspection effort.
- Consistent surface quality analysis.
- Improved identification of color variations.
- Clear visualization of inspection results.

Future work can include collecting feedback from manufacturing quality-control engineers and industrial inspection professionals.

Appendix I – Publications / Patents / Competitions / Product Outcomes

At the current stage, no publications, patents, competitions, or commercial product outcomes have been claimed for this project.

The potential future outcomes include:

- Research publication on integrated metal surface segmentation.
- Development of an industrial inspection prototype.
- Participation in computer vision or AI project competitions.
- Extension into a real-time quality-control application.

- Potential intellectual property development after further implementation and validation.

Appendix J – Individual Contribution Evidence

Student Name	Register Number	Primary Contribution
M. Maruthi	192425257	Project coordination, system design and U-Net segmentation
Anvesh Chandra Reddy	192472303	Image preprocessing and dataset preparation
Krishna Tejesh	192411200	K-Means clustering and color analysis
A. Archishman	192425341	Visualization and performance evaluation
K. Subrahmanya Venkat Sai	192472220	System integration, testing and documentation

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)

Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)